176100

PORPHYRIA CUTANEA TARDA

Alternative titles; symbols

PCT
PORPHYRIA CUTANEA TARDA, TYPE II
PCT, TYPE II
PCT, 'FAMILIAL' TYPE
PORPHYRIA, HEPATOCUTANEOUS TYPE
UROPORPHYRINOGEN DECARBOXYLASE DEFICIENCY
UROD DEFICIENCY

Other entities represented in this entry:

PORPHYRIA, HEPATOERYTHROPOIETIC, INCLUDED; HEP, INCLUDED

SNOMEDCT: 111386004, 276262000, 59229005, 61860000; **ICD10CM:** E80.1; **ORPHA:** 101330, 443062, 95159; **DO:** 3132; **MONDO:** 0008296;

Phenotype-Gene Relationships

Location	Phenotype	Phenotype MIM number	Inheritance	Phenotype mapping key	Gene/Locus	Gene/Locus MIM number
1p34.1	Porphyria cutanea tarda	176100	Autosomal dominant; Autosomal recessive	3	UROD	613521
1p34.1	Porphyria, hepatoerythropoietic	176100	Autosomal dominant; Autosomal recessive	3	UROD	613521

TEXT

A number sign (#) is used with this entry because porphyria cutanea tarda type II, or familial PCT, is caused by heterozygous mutation in the gene encoding uroporphyrinogen decarboxylase (UROD; 613521). Hepatoerythropoietic porphyria (HEP) is caused by homozygous or compound heterozygous mutation in the UROD gene.

Description

Porphyria cutanea tarda (PCT) is characterized by light-sensitive dermatitis and the excretion of large amounts of uroporphyrin in urine (Elder et al., 1980).

De Verneuil et al. (1978) and others classified porphyria cutanea tarda, the most common type of porphyria, into 2 types: type I (176090), or 'sporadic' type, associated with approximately 50% level of uroporphyrinogen decarboxylase (UROD) in liver (Elder et al., 1978; Felsher et al., 1982), and type II, or 'familial' type, characterized by 50% deficient activity of the same enzyme in many tissues (Kushner et al., 1976; Elder et al., 1980).

PCT type II is an autosomal dominant disorder with low penetrance and constitutes about 20% of cases of PCT. Recognized exacerbating factors of PCT include iron overload, excessive use of alcohol, exposure to polyhalogenated aromatic chemicals, exposure to estrogens, chronic viral hepatitis C, HIV infections, and mutation in the HFE gene (613609) that are responsible for hereditary hemochromatosis (235200) (review by Lambrecht et al., 2007).

Clinical Features

Onset of light-sensitive dermatitis in later adult life, associated with the excretion of large amounts of uroporphyrin in urine, characterizes porphyria cutanea tarda, which was so named by Waldenstrom (1937). On areas of skin exposed to sunlight, especially the face, ears, and backs of the hands, chronic ulcerating lesions commence as blisters, and the skin may also be mechanically fragile (Grossman et al., 1979). Hyperpigmentation and hypertrichosis also occur. Acute neuropathic episodes do not occur in this form of porphyria. Onset is often associated with alcoholism, and occasionally with exposure to other agents, such as estrogens. Iron overload is frequently present, and may be associated, coincidentally or causally, with varying degrees of liver damage or fibrosis; liver histology may be characteristic (Cortes et al., 1980). On biopsy, liver parenchyma cells are also loaded with porphyrins and fluoresce bright red in ultraviolet light. The skin lesions are distinctly related to circulating porphyrins (Holti et al., 1958).

Malina and Lim (1988) described a 29-year-old woman who first presented with blisters and erosions on the dorsum of the fingers and hands bilaterally 3 weeks after delivery of her second child. The diagnosis of PCT was established enzymatically and by porphyrin studies. Reduced red cell UROD activity was found also in the newborn child and in the patient's mother.

Classic congenital erythropoietic porphyria (263700) is due to deficiency of uroporphyrinogen III cosynthase. Kushner et al. (1982) described a remarkable 51-year-old man with congenital erythropoietic porphyria (Gunther disease), first manifested in infancy with eventual development of mutilating skin photosensitivity. The morphologic features of dyserythropoietic bone marrow cells, studied by light and electron microscopy, were identical to those found in congenital dyserythropoietic anemia type I (224120); such had been described before in Gunther disease. A red-orange nuclear fluorescence is not seen in type I dyserythropoietic anemia. The patient of Kushner et al. (1982) showed massive porphyrinuria, but the pattern of porphyrin excretion was atypical for classic Gunther disease: hepta-carboxyl (7-COOH) porphyrin was the major urine porphyrin, much uroporphyrin was present, and both were predominantly of the isomer III type. Erythrocyte uroporphyrinogen III cosynthase activity was normal, but uroporphyrinogen decarboxylase activity was 50% of normal. Two sons showed equally subnormal uroporphyrinogen decarboxylase activity. It was the opinion of the authors that their 51-year-old patient had 2 genetic diseases--uroporphyrinogen decarboxylase deficiency (a heterozygous state) and type I congenital dyserythropoietic anemia (a presumably homozygous state). With coexisting hepatic siderosis, heterozygous uroporphyrinogen decarboxylase deficiency leads to porphyria cutanea tarda. Homozygosity for a deficiency gene leads to hepatoerythropoietic porphyria. Thus, Gunther disease can have more than one cause. Two other reported patients with clinically typical congenital erythropoietic porphyria, but with a pattern of urinary porphyrin excretion similar to porphyria cutanea tarda, were referenced by Kushner et al. (1982).

Hepatoerythropoietic Porphyria

Hepatoerythropoietic porphyria (HEP) is a severe, autosomal recessive form of cutaneous porphyria that presents in infancy and is characterized biochemically by excessive excretion of acetate-substituted porphyrins and accumulation of protoporphyrin in erythrocytes (Hofstad et al., 1973; Simon et al., 1977; Czarnecki, 1980). As in porphyria cutanea tarda, uroporphyrinogen decarboxylase is deficient. However, the enzyme level is very low (7-8%) in erythrocytes and cultured skin fibroblasts, leading Elder et al. (1981) to propose that HEP is the homozygous state for porphyria cutanea tarda.

De Verneuil et al. (1984) brought to 9 the number of known cases of HEP and confirmed that these patients are homozygous for mutations in the same gene that causes PCT. The patients of de Verneuil et al. (1984) were twin daughters of a Tunisian couple related as second cousins. Both parents, although asymptomatic, showed intermediate levels of enzymatic and immunoreactive URO decarboxylase. The twins were CRM-negative, in contrast to previously reported homozygous patients.

Toback et al. (1987) described a man with relatively mild hepatoerythropoietic porphyria and concluded that the man was a homozygote since both of his parents and his 3 children, all of whom were asymptomatic, showed moderate deficiency of UROD. They concluded that the relative mildness of the clinical symptoms in the proband was probably related to the level of residual enzyme activity and that the genetic defect in UROD in this disorder can be heterogeneous.

Fujimoto and Brazil (1992) reported a 23-year-old woman thought to represent the 18th instance of HEP reported worldwide. She had photosensitive skin of early onset, hypertrichosis, and severe scleroderma-like lesions of the hands.

PCT 'Phenocopy'

A syndrome similar to PCT, a 'phenocopy,' is caused by toxic exposure to certain organic chemicals such as hexachlorobenzene, as in the epidemic caused by contaminated seed wheat in Turkey (Cam and Nigogosyan, 1963; Dean, 1972) and by occupational exposure to chlorinated hydrocarbons (Bleiberg et al., 1964).

Pathogenesis

Felsher et al. (1982) concluded that reduced hepatic uroporphyrinogen decarboxylase activity is a specific and intrinsic hepatic defect in PCT, but modulation of uroporphyrinogen synthesis by extrinsic factors is required for full biochemical expression of the disease.

Biochemical Features

Reduced liver and red cell uroporphyrinogen decarboxylase activity has been reported in familial (Kushner et al., 1976; Lehr and Doss, 1981) and sporadic cases of porphyria cutanea tarda (Elder et al., 1978; Felsher et al., 1978). Impaired activity of this enzyme step in heme synthesis in liver could possibly explain resulting 'overflow' of uroporphyrin. Hepatic uroporphyrinogen decarboxylase activity was reduced to approximately 50% of normal levels in 17 cases of porphyria cutanea tarda and reduced levels persisted after hepatic iron overload was relieved by phlebotomy (Felsher et al., 1982). Elder et al. (1978) found normal levels of enzyme in red cells and fibroblasts. In assays of UROD activity in red cells, de Verneuil et al. (1978) found 50% levels of uroporphyrinogen decarboxylase in persons with familial porphyria cutanea, but normal enzyme levels in sporadic cases.

In hemolysates from 7 unrelated patients with familial PCT, Elder et al. (1983) found that immunoreactive uroporphyrinogen decarboxylase was decreased (average 51% of normal) to the same extent as catalytic activity (average 56% of normal), whereas in 6 sporadic cases both measurements were normal. The failure to find evidence of CRM+ mutations among the familial cases suggested to Elder et al. (1983) that a simple immunoelectrophoretic method can be used for routine diagnosis.

Using a UROD cDNA probe in Northern blot analysis, Hansen et al. (1988) found no difference in the levels of UROD mRNA between affected individuals and their normal relatives.

Inheritance

Most cases of PCT are sporadic and are more common in men than women, but familial cases have been described frequently, and apparent autosomal dominant segregation of the disorder has been reported (Holti et al., 1958; Ziprkowski et al., 1966; Topi and D'Alessandro Gandolfo, 1977; Benedetto et al., 1978).

Although it is unusual for an enzyme deficiency to produce symptoms in the heterozygous state, i.e., in single gene dose, this is also the pattern in other types of genetic porphyrias (e.g., 121300, 176000, 176200, 177000). It seems likely that a reduced level of activity of uroporphyrinogen decarboxylase may segregate as an autosomal dominant trait, but that additional environmental factors are required for manifestation of the disorder; iron overload may have a direct metabolic role (Kushner et al., 1972; Kushner, 1982).

Blekkenhorst et al. (1979) suggested that 2 forms of PCT exist: a rare familial form and a relatively common idiosyncratic form occurring sporadically as an unusual accompaniment of common hepatic disorders such as alcohol-associated liver disease.

Hepatoerythropoietic porphyria (HEP) is an autosomal recessive trait (de Verneuil et al., 1984).

Population Genetics

The incidence of PCT varies from approximately 1 in 25,000 in the United States to approximately 1 in 5,000 in the Czech Republic and Slovakia (review by Lambrecht et al., 2007).

PCT is common in the Bantu races in South Africa in association with iron overload (Barnes, 1955).

Clinical Management

Treatment is directed first to reducing iron overload by regular phlebotomy, as in the management of hemochromatosis (Epstein and Redeker, 1968; Ramsay et al., 1974; Grossman et al., 1979). Porphyrin excretion diminishes, and in many patients skin lesions disappear. When this is ineffective or when a more rapid effect is desired, oral chloroquine therapy usually induces rapid remission (Taljaard et al., 1972; Kowertz, 1973). It may also cause a transient increase in porphyrin excretion, sometimes associated with evidence of acute liver damage (Vogler et al., 1970). Remission is sustained while chloroquine is continued in regular low doses.

Several cases of porphyria cutanea tarda have been described in patients on maintenance hemodialysis for chronic renal failure (e.g., Poh-Fitzpatrick et al., 1978). The cause is thought to be insufficient removal of porphyrins through the hemodialysis membrane which leads to markedly increased levels of plasma porphyrins with resulting severe and mutilating skin lesions. The treatment of the disorder is very difficult because chloroquine is ineffective and the anemia that accompanies chronic renal failure contraindicates venesection therapy. Praga et al. (1987) found that deferoxamine was effective therapy in a patient in whom there was evidence of iron overload due to multiple blood transfusions.

Molecular Genetics

Using hybridization probes for the UROD gene in the study of genomic DNA from patients with familial PCT, Hansen et al. (1988) could not identify any major deletions, rearrangements, or restriction fragment length polymorphisms.

In the UROD cDNA from a patient with familial PCT, Garey et al. (1989) demonstrated a gly-to-val substitution at amino acid position 281 (G281V; 613521.0001). The mutation was not detected in affected persons from 7 other PCT pedigrees with an autosomal dominant pattern. They showed that the UROD protein in the patient with the identified mutation had a greatly shortened half-life, both in vitro and in vivo (assuming, as these workers did, that one can call the findings in cultured lymphocytes an 'in vivo' observation). Hepatoerythropoietic porphyria results from a different nucleotide substitution in the same codon (G281E; 613521.0002). The UROD protein resulting from the G281E mutation also has a decreased half-life, but not so severely decreased as in the case of the G281V mutation. Garey et al. (1989) suggested that the former mutation may be so severe in the homozygous state that it is lethal to the embryo; PCT can result in the heterozygote for the first mutation, but only the homozygote for the milder mutation expresses itself (as HEP). Garey et al. (1989) pointed out that familial PCT is relatively common, but only 16 cases of HEP have been described to date.

Using a cDNA clone for the UROD gene, de Verneuil et al. (1986) studied DNA from 2 homozygous patients, offspring of consanguineous parents, who suffered from HEP. They could detect neither deletions nor rearrangements in the UROD gene. Synthesis, processing, and cell-free translation of the specific transcripts appeared to be normal. The half-life of the abnormal protein was 12 times shorter than that of the normal enzyme. Thus, rapid degradation in vivo is the probable basis of the enzyme deficiency. Study of homozygous patients avoided the difficulties of studying the enzyme defect in the heterozygous PCT where both normal and abnormal protein is present. The authors suggested that use of oligonucleotide probes complementary to the normal and mutant sequences could allow them to determine if the mutation in familial PCT is the same as that in HEP; in other words, whether HEP is indeed the homozygous state of PCT.

In a Spanish family, Moran-Jimenez et al. (1996) found homozygosity for the G281E (613521.0001) mutation as the cause of HEP. A paternal uncle of the proband developed clinically overt porphyria cutanea tarda as an adult and proved to be heterozygous for the G281E mutation.

Mendez et al. (1998) sequenced the entire UROD gene, and developed a long-range PCR method to amplify the entire gene for mutation analysis. Four missense mutations (M165R, 613521.0009; L195F, 613521.0010; N304K, 613521.0011; and R332H, 613521.0012), a microinsertion, a deletion, and a novel exonic splicing defect were identified. Expression of the L195F, N304K, and R332H polypeptides revealed significant residual activity, whereas RT-PCR and sequencing demonstrated that the E314E (613521.0008) lesion caused abnormal splicing and exon 9 skipping. Screening of 9 familial PCT probands revealed that 4 (44%) were heterozygous or homozygous for the common hemochromatosis mutations, which suggested that iron overload may predispose to clinical expression. However, there was no clear correlation between the severity of familial PCT and the UROD and/or hemochromatosis genotypes. Presymptomatic molecular diagnosis should now be possible, permitting counseling to enable family members to avoid disease-precipitating factors.

Role of Mutations in the HFE Gene

An association between PCT and HLA-linked hereditary hemochromatosis (HFE; 235200) was suggested by Kushner et al. (1985), but disputed by Beaumont et al. (1986). Santos et al. (1997) assessed the role of HFE (613609) mutations in PCT by an allelic-association study between PCT and the mutations identified in hemochromatosis. They studied 15 unselected, unrelated patients with PCT being treated with regular phlebotomy. The controls were 23 anonymous blood donors and 71 patients with hereditary hemochromatosis. The cys282-to-tyr mutation (C282Y; 613609.0001) was found in 83% of 142 hereditary hemochromatosis chromosomes, 47% of 30 PCT chromosomes, and 9% of 46 normal blood donor chromosomes. Santos et al. (1997) concluded that the hemochromatosis gene contributes to the pathogenesis of PCT. They suggested that all first-degree relatives of patients with PCT should be screened for hereditary hemochromatosis. PCT can be viewed as having a digenic basis.

Ivanova et al. (1999) found the C282Y mutation of the HFE gene in only 1 of 48 PCT patients (2.1%). This individual was heterozygous for the mutation. The mutation was found in none of 100 healthy Bulgarian subjects. This indicates a very low frequency of the C282Y mutation in Bulgaria. A similarly low frequency of HFE mutations was found in Japanese cases of PCT and in Japanese patients generally, leading Furuyama et al. (1999) to suggest that abnormal iron metabolism associated with PCT in Japanese patients occurs by a mechanism unrelated to HFE gene mutations.

Brady et al. (2000) investigated the relationship between age of onset of skin lesions and mutations (C282Y, 613609.0001; H63D, 613609.0002) in the hemochromatosis gene in 19 familial and 65 sporadic porphyria cutanea tarda patients. Familial porphyria cutanea tarda was identified by mutation analysis of the uroporphyrinogen decarboxylase gene. Five previously described and 8 novel mutations were identified. Homozygosity for the C282Y hemochromatosis mutation was associated with an earlier onset of skin lesions in both familial and sporadic porphyria cutanea tarda, the effect being more marked in familial porphyria cutanea tarda where anticipation was demonstrated in family studies. Analysis of the frequencies of hemochromatosis genotypes in each type of porphyria cutanea tarda indicated that C282Y homozygosity is an important susceptibility factor in both types but suggested that heterozygosity for this mutation has much less effect on the development of the disease.

Dereure et al. (2001) evaluated 36 consecutive patients with either sporadic or familial PCT for the presence of the 3 main mutations of the HFE gene and identification of the transferrin receptor alleles. Seven patients (19%) showed heterozygous C282Y (613609.0001) mutation, but no C282Y homozygote was present; 5 patients (14%) carried homozygous H63D (613609.0002) mutation, while 8 (22%) were heterozygous for this mutation. One patient was heterozygous for the S65C (613609.0003) mutation (3%). Iron parameters demonstrated overload in all patients, without a clear difference between patients with and without deleterious mutations of the HFE gene. Infection by hepatitis C virus was documented in 20 patients (56%), and was significantly less frequent in patients with deleterious HFE mutations. The profile of transferrin receptor alleles in PCT patients did not show significant variation compared with the general population. Dereure et al. (2001) concluded that there is a high frequency of HFE mutations in patients with PCT and that HFE gene abnormalities might play a significant part in the PCT pathomechanism, probably through iron overload; by contrast, transferrin receptor polymorphisms do not appear to play a significant part in iron overload in PCT.

Stolzel et al. (2003) retrospectively analyzed 62 German PCT patients exclusively treated with low-dose chloroquine to determine whether HFE mutations C282Y (613609.0001) and H63D (613609.0002) influenced the clinical response, urinary porphyrin excretion, liver enzyme activities, and serum iron markers. Chloroquine therapy was accompanied by clinical remission and reduced urinary porphyrin excretion in the 24 patients (39%) with HFE wildtype as well as in 35 HFE heterozygous patients with PCT (56%). Decreases of serum iron markers following chloroquine therapy were limited to patients with PCT and HFE wildtype. All 3 patients homozygous for the C282Y mutation (5%) had high serum iron, ferritin, and transferrin saturation and failed to respond to chloroquine treatment. Stolzel et al. (2003) concluded that the therapeutic response to chloroquine was not compromised by C282Y heterozygosity and compound heterozygosity of HFE mutations. However, because HFE C282Y homozygotes did not respond to chloroquine and a decrease in serum iron concentration was limited to patients with PCT and HFE wildtype, phlebotomy should be first-line therapy in patients with PCT and HFE mutations.

Role of Mutations in the CYP12A Gene

Individuals with PCT are believed to be genetically predisposed to development of clinically overt disease through mutations and polymorphisms in particular genes in response to precipitating factors. Christiansen et al. (2000) examined a group of Danish patients with PCT for the presence of a C/A polymorphism in intron 1 of CYP1A2 (124060). The results demonstrated that the frequency of the highly inducible A/A genotype is increased in both familial and sporadic PCT. This suggested that inheritance of this genotype is a susceptibility factor for PCT.

Animal Model

The zebrafish mutant 'yquem' shows a photosensitive porphyria syndrome. Wang et al. (1998) showed that the porphyric phenotype is due to an inherited homozygous mutation in the UROD gene. Thus, the zebrafish mutant represented the first genetically 'accurate' model of hepatoerythropoietic porphyria; Wang et al. (1998) suggested that the model would be useful for studying the pathogenesis of UROD deficiency and evaluating gene therapy vectors. Wang et al. (1998) rescued the mutant phenotype by transient and germline expression of the wildtype allele.

Most heterozygotes for UROD mutations do not express a porphyric phenotype unless hepatic siderosis is present. Mutations in the hemochromatosis gene are frequently found when the porphyric phenotype is expressed in the heterozygote. Phillips et al. (2001) used homologous recombination to disrupt 1 allele of the murine Urod gene. Urod +/- mice had half-wildtype UROD protein and enzymatic activity in all tissues but did not accumulate hepatic porphyrins, indicating that half-normal UROD activity is not rate limiting. When Urod +/- mice were injected with iron-dextran and given drinking water containing delta-aminolevulinic acid (ALA) for 21 days, hepatic porphyrins accumulated, and hepatic UROD activity was reduced to 20% of weight. Phillips et al. (2001) also bred mice homozygous for the HFE gene disruption (Hfe -/-) to Urod +/- mice, generating mice with the heterozygous Urod genotype and the homozygous null Hfe

genotype. These animals developed a porphyric phenotype by 14 weeks of age without ALA supplementation, and UROD activity was reduced to 14% of weight. These data indicated that iron overload alone is sufficient to reduce UROD activity to rate-limiting levels in heterozygous Urod mice. Thus these mice serve as an excellent model of familial PCT and afford the opportunity to define the mechanism by which iron influences UROD activity.

See Also:

Day et al. (1982); Hansen et al. (1988); Romana et al. (1991); Romeo (1977)

REFERENCES

1. Barnes, H. D. **Porphyria in the Bantu races on the Witwatersrand.** S. Afr. Med. J. 29: 781-784, 1955. [PubMed: 13256018]
2. Beaumont, C., Nordmann, Y., Fauchet, R. **HLA-linked hemochromatosis and sporadic porphyria cutanea tarda. (Letter)** Gastroenterology 90: 800-801, 1986. [PubMed: 3943706] [Full Text: https://doi.org/10.1016/0016-5085(86)91155-8]
3. Benedetto, A. V., Kushner, J. P., Taylor, J. S. **Porphyria cutanea tarda in three generations of a single family.** New Eng. J. Med. 298: 358-361, 1978. [PubMed: 622106] [Full Text: https://doi.org/10.1056/NEJM197802162980702]
4. Bleiberg, J., Wallen, M., Brodkin, R., Applebaum, I. **Industrially acquired porphyria.** Arch. Derm. 89: 793-797, 1964. [PubMed: 14164959] [Full Text: https://doi.org/10.1001/archderm.1964.01590300021006]
5. Blekkenhorst, G. H., Day, R. S., Eales, L. **Two forms of porphyria cutanea tarda? (Letter)** New Eng. J. Med. 300: 93, 1979. [PubMed: 758588] [Full Text: https://doi.org/10.1056/NEJM197901113000216]
6. Brady, J. J., Jackson, H. A., Roberts, A. G., Morgan, R. R., Whatley, S. D., Rowlands, G. L., Darby, C., Shudell, E., Watson, R., Paiker, J., Worwood, M. W., Elder, G. H. **Co-inheritance of mutations in the uroporphyrinogen decarboxylase and hemochromatosis genes accelerates the onset of porphyria cutanea tarda.** J. Invest. Derm. 115: 868-874, 2000. [PubMed: 11069625] [Full Text: https://doi.org/10.1046/j.1523-1747.2000.00148.x]
7. Cam, C., Nigogosyan, G. **Acquired toxic porphyria cutanea tarda due to hexachlorobenzene.** JAMA 183: 88-91, 1963. [PubMed: 14018209]
8. Christiansen, L., Bygum, A., Jensen, A., Thomsen, K., Brandrup, F., Horder, M., Petersen, N. E. **Association between CYP1A2 polymorphism and susceptibility to porphyria cutanea tarda.** Hum. Genet. 107: 612-614, 2000. [PubMed: 11153915] [Full Text: https://doi.org/10.1007/s004390000415]
9. Cortes, J. M., Oliva, H., Paradinas, F. J., Hernandez-Guio, C. **The pathology of the liver in porphyria cutanea tarda.** Histopathology 4: 471-485, 1980. [PubMed: 6253381] [Full Text: https://doi.org/10.1111/j.1365-2559.1980.tb02942.x]
10. Czarnecki, D. B. **Hepatoerythropoietic porphyria.** Arch. Derm. 116: 307-311, 1980. [PubMed: 7369748]
11. Day, R. S., Eales, L., Meissner, D. **Coexistent variegate porphyria and porphyria cutanea tarda.** New Eng. J. Med. 307: 36-41, 1982. [PubMed: 7078623] [Full Text: https://doi.org/10.1056/NEJM198207013070107]
12. de Verneuil, H., Aitken, G., Nordmann, Y. **Familial and sporadic porphyria cutanea: two different diseases.** Hum. Genet. 44: 145-151, 1978. [PubMed: 730158] [Full Text: https://doi.org/10.1007/BF00295407]
13. de Verneuil, H., Beaumont, C., Deybach, J.-C., Nordmann, Y., Sfar, Z., Kastally, R. **Enzymatic and immunological studies of uroporphyrinogen decarboxylase in familial porphyria cutanea tarda and hepatoerythropoietic porphyria.** Am. J. Hum. Genet. 36: 613-622, 1984. [PubMed: 6375356]
14. de Verneuil, H., Grandchamp, B., Beaumont, C., Picat, C., Nordmann, Y. **Uroporphyrinogen decarboxylase structural mutant (gly-281-to-glu) in a case of porphyria.** Science 234: 732-734, 1986. [PubMed: 3775362] [Full Text: https://doi.org/10.1126/science.3775362]
15. Dean, G. **The Porphyrias. A Story of Inheritance and Environment. (2nd ed.)** Philadelphia: J. B. Lippincott (pub.) 1972.
16. Dereure, O., Aguilar-Martinez, P., Bessis, D., Perney, P., Vallat, C., Guillot, B., Blanc, F., Guilhou, J.-J. **HFE mutations and transferrin receptor polymorphism analysis in porphyria cutanea tarda: a prospective study of 36 cases from southern France.** Brit. J. Derm. 144: 533-539, 2001. [PubMed: 11260010] [Full Text: https://doi.org/10.1046/j.1365-2133.2001.04079.x]
17. Elder, G. H., Lee, G. B., Tovey, J. A. **Decreased activity of hepatic uroporphyrinogen decarboxylase in sporadic porphyria cutanea tarda.** New Eng. J. Med. 299: 274-278, 1978. [PubMed: 661926] [Full Text: https://doi.org/10.1056/NEJM197808102990603]
18. Elder, G. H., Sheppard, D. M., de Salamanca, R. E., Olmos, A. **Identification of two types of porphyria cutanea tarda by measurement of erythrocyte uroporphyrinogen decarboxylase.** Clin. Sci. (Lond.) 58: 477-484, 1980. [PubMed: 7428280] [Full Text: https://doi.org/10.1042/cs0580477]
19. Elder, G. H., Sheppard, D. M., Tovey, J. A., Urquhart, A. J. **Immunoreactive uroporphyrinogen decarboxylase in porphyria cutanea tarda.** Lancet 321: 1301-1304, 1983. Note: Originally Volume I. [PubMed: 6134095] [Full Text: https://doi.org/10.1016/s0140-6736(83)92414-5]
20. Elder, G. H., Smith, S. G., Herrero, C., Mascaro, J. M., Lecha, M., Muniesa, A. M., Czarnecki, D. B., Brenan, J., Poulos, V., de Salamanca, R. E. **Hepatoerythropoietic porphyria: a new uroporphyrinogen decarboxylase defect or homozygous porphyria cutanea tarda?** Lancet 317: 916-919, 1981. Note: Originally Volume I. [PubMed: 6112327] [Full Text: https://doi.org/10.1016/s0140-6736(81)91615-9]

21. Epstein, J. H., Redeker, A. G. **Porphyria cutanea tarda: a study of the effect of phlebotomy.** New Eng. J. Med. 279: 1301-1304, 1968. [PubMed: 5697519] [Full Text: https://doi.org/10.1056/NEJM196812122792402]

22. Felsher, B. F., Carpio, N. M., Engleking, D. W., Nunn, A. T. **Decreased hepatic uroporphyrinogen decarboxylase activity in porphyria cutanea tarda.** New Eng. J. Med. 306: 766-769, 1982. [PubMed: 7062951] [Full Text: https://doi.org/10.1056/NEJM198204013061302]

23. Felsher, B. F., Norris, M. E., Shih, J. C. **Red-cell uroporphyrinogen decarboxylase activity in porphyria cutanea tarda and in other forms of porphyria.** New Eng. J. Med. 299: 1095-1098, 1978. [PubMed: 703786] [Full Text: https://doi.org/10.1056/NEJM197811162992002]

24. Fujimoto, A., Brazil, J. L. **Hepatoerythropoietic porphyria in a woman with short stature and deformed hands.** Am. J. Med. Genet. 44: 496-499, 1992. [PubMed: 1442894] [Full Text: https://doi.org/10.1002/ajmg.1320440423]

25. Furuyama, K., Kondo, M., Hirata, K., Fujita, H., Sassa, S. **Extremely rare association of HFE mutations with porphyria cutanea tarda in Japanese patients. (Letter)** Hepatology 30: 1532-1533, 1999. [PubMed: 10610355] [Full Text: https://doi.org/10.1002/hep.510300627]

26. Garey, J. R., Hansen, J. L., Harrison, L. M., Kennedy, J. B., Kushner, J. P. **A point mutation in the coding region of uroporphyrinogen decarboxylase associated with familial porphyria cutanea tarda.** Blood 73: 892-895, 1989. [PubMed: 2920211]

27. Grossman, M. E., Bickers, D. R., Poh-Fitzpatrick, M. B., Delco, V. A., Harber, L. C. **Porphyria cutanea tarda: clinical features and laboratory findings in 40 patients.** Am. J. Med. 67: 277-286, 1979. [PubMed: 463934] [Full Text: https://doi.org/10.1016/0002-9343(79)90403-0]

28. Hansen, J. L., O'Connell, P. O., Romana, M., Romeo, P.-H., Kushner, J. P. **Familial porphyria cutanea tarda: hybridization analysis of the uroporphyrinogen decarboxylase locus.** Hum. Hered. 38: 283-286, 1988. [PubMed: 2906904] [Full Text: https://doi.org/10.1159/000153800]

29. Hansen, J. L., Pryor, M. A., Kennedy, J. B., Kushner, J. P. **Steady-state levels of uroporphyrinogen decarboxylase mRNA in lymphoblastoid cell lines from patients with familial porphyria cutanea tarda and their relatives.** Am. J. Hum. Genet. 42: 847-853, 1988. [PubMed: 3369447]

30. Hofstad, F., Seip, M., Eriksen, L. **Congenital erythropoietic porphyria with a hitherto undescribed porphyrin pattern.** Acta Paediat. Scand. 62: 380-384, 1973. [PubMed: 4729688] [Full Text: https://doi.org/10.1111/j.1651-2227.1973.tb08123.x]

31. Holti, G., Rimington, C., Tate, B. C., Thomas, G. **An investigation of 'porphyria cutanea tarda'.** Quart. J. Med. 27: 1-19, 1958. [PubMed: 13506007]

32. Ivanova, A., von Ahsen, N., Adjarov, D., Krastev, Z., Oellerich, M., Wieland, E. **C282Y and H63D mutations in the HFE gene are not associated with porphyria cutanea tarda in Bulgaria. (Letter)** Hepatology 30: 1531-1532, 1999. [PubMed: 10610354] [Full Text: https://doi.org/10.1002/hep.510300626]

33. Kowertz, M. J. **The therapeutic effect of chloroquine.** JAMA 223: 515-519, 1973. [PubMed: 4739135]

34. Kushner, J. P., Barbuto, A. J., Lee, G. R. **An inherited enzymatic defect in porphyria cutanea tarda: decreased uroporphyrinogen decarboxylase activity.** J. Clin. Invest. 58: 1089-1097, 1976. [PubMed: 993332] [Full Text: https://doi.org/10.1172/JCI108560]

35. Kushner, J. P., Edwards, C. Q., Dadone, M. M., Skolnick, M. H. **Heterozygosity for HLA-linked hemochromatosis as a likely cause of the hepatic siderosis associated with sporadic porphyria cutanea tarda.** Gastroenterology 88: 1232-1238, 1985. [PubMed: 3979748] [Full Text: https://doi.org/10.1016/s0016-5085(85)80084-6]

36. Kushner, J. P., Lee, G. R., Nacht, S. **The role of iron in the pathogenesis of porphyria cutanea tarda: an in vitro model.** J. Clin. Invest. 51: 3044-3051, 1972. [PubMed: 4640947] [Full Text: https://doi.org/10.1172/JCI107131]

37. Kushner, J. P., Pimstone, N. R., Kjeldsberg, C. R., Pryor, M. A., Huntley, A. **Congenital erythropoietic porphyria, diminished activity of uroporphyrinogen decarboxylase and dyserythropoiesis.** Blood 59: 725-737, 1982. [PubMed: 7059676]

38. Kushner, J. P. **The enzymatic defect in porphyria cutanea tarda. (Editorial)** New Eng. J. Med. 306: 799-800, 1982. [PubMed: 7062954] [Full Text: https://doi.org/10.1056/NEJM198204013061309]

39. Lambrecht, R. W., Thapar, M., Bonkovsky, H. L. **Genetic aspects of porphyria cutanea tarda.** Semin. Liver Dis. 27: 99-108, 2007. [PubMed: 17295179] [Full Text: https://doi.org/10.1055/s-2006-960173]

40. Lehr, P. A., Doss, M. **Chronishe hepatische porphyrie mit Uroporphyrinogen-Decarboxylase-Defekt in vier Generationen.** Dtsch. Med. Wschr. 106: 241-245, 1981. [PubMed: 7472211] [Full Text: https://doi.org/10.1055/s-2008-1070295]

41. Malina, L., Lim, C. K. **Manifestation of familial porphyria cutanea tarda after childbirth.** Brit. J. Derm. 118: 243-245, 1988. [PubMed: 3348969] [Full Text: https://doi.org/10.1111/j.1365-2133.1988.tb01781.x]

42. Mendez, M., Sorkin, L., Rossetti, M. V., Astrin, K. H., Batlle, A. M. del C., Parera, V. E., Aizencang, G., Desnick, R. J. **Familial porphyria cutanea tarda: characterization of seven novel uroporphyrinogen decarboxylase mutations and frequency of common hemochromatosis alleles.** Am. J. Hum. Genet. 63: 1363-1375, 1998. [PubMed: 9792863] [Full Text: https://doi.org/10.1086/302119]

43. Moran-Jimenez, M. J., Ged, C., Romana, M., Enriquez de Salamanca, R., Taieb, A., Topi, G., D'Alessandro, L., de Verneuil, H. **Uroporphyrinogen decarboxylase: complete human gene sequence and molecular study of three families with hepatoerythropoietic porphyria.** Am. J. Hum. Genet. 58: 712-721, 1996. [PubMed: 8644733]

44. Phillips, J. D., Jackson, L. K., Bunting, M., Franklin, M. R., Thomas, K. R., Levy, J. E., Andrews, N. C., Kushner, J. P. **A mouse model of familial porphyria cutanea tarda.** Proc. Nat. Acad. Sci. 98: 259-264, 2001. [PubMed: 11134514] [Full Text: https://doi.org/10.1073/pnas.98.1.259]

45. Poh-Fitzpatrick, M. B., Bellet, N., DeLeo, V. A., Grossman, M. E., Bickers, D. R. **Porphyria cutanea tarda in two patients treated with hemodialysis for chronic renal failure.** New Eng. J. Med. 299: 292-294, 1978. [PubMed: 661929] [Full Text: https://doi.org/10.1056/NEJM197808102990607]

46. Praga, M., Enriquez de Salamanca, R. E., Andres, A., Nieto, J., Oliet, A., Perpina, J., Morales, J. M. **Treatment of hemodialysis-related porphyria cutanea tarda with deferoxamine. (Letter)** New Eng. J. Med. 316: 547-548, 1987. [PubMed: 3808000]

47. Ramsay, C. A., Magnus, I. A., Turnbull, A., Baker, H. **The treatment of porphyria cutanea tarda by venesection.** Quart. J. Med. 43: 1-24, 1974. [PubMed: 4822971]

48. Romana, M., Grandchamp, B., Dubart, A., Amselem, S., Chabret, C., Nordmann, Y., Goossens, M., Romeo, P.-H. **Identification of a new mutation responsible for hepatoerythropoietic porphyria.** Europ. J. Clin. Invest. 21: 225-229, 1991. [PubMed: 1905636] [Full Text: https://doi.org/10.1111/j.1365-2362.1991.tb01814.x]

49. Romeo, G. **Enzymatic defects of hereditary porphyrias.** Hum. Genet. 39: 261-276, 1977. [PubMed: 340378] [Full Text: https://doi.org/10.1007/BF00295419]

50. Santos, M., Clevers, H. C., Marx, J. J. M. **Mutations of the hereditary hemochromatosis candidate gene HLA-H in porphyria cutanea tarda. (Letter)** New Eng. J. Med. 336: 1327-1328, 1997. [PubMed: 9132598] [Full Text: https://doi.org/10.1056/NEJM199705013361817]

51. Simon, N., Berko, G., Schneider, I. **Hepato-erythropoietic porphyria presenting as scleroderma and acrosclerosis in a sibling pair.** Brit. J. Derm. 96: 663-668, 1977. [PubMed: 871403] [Full Text: https://doi.org/10.1111/j.1365-2133.1977.tb05212.x]

52. Stolzel, U., Kostler, E., Schuppan, D., Richter, M., Wollina, U., Doss, M. O., Wittekind, C., Tannapfel, A. **Hemochromatosis (HFE) gene mutations and response to chloroquine in porphyria cutanea tarda.** Arch. Derm. 139: 309-313, 2003. [PubMed: 12622622] [Full Text: https://doi.org/10.1001/archderm.139.3.309]

53. Taljaard, J. F., Shanley, B. C., Stewart-Wynne, E., Deppe, W. M., Joubert, S. M. **Studies on low dose chloroquine therapy and the action of chloroquine in symptomatic porphyria.** Brit. J. Derm. 87: 261-269, 1972. [PubMed: 5080345] [Full Text: https://doi.org/10.1111/j.1365-2133.1972.tb00316.x]

54. Toback, A. C., Sassa, S., Poh-Fitzpatrick, M. B., Schechter, J., Zaider, E., Harber, L. C., Kappas, A. **Hepatoerythropoietic porphyria: clinical, biochemical, and enzymatic studies in a three-generation family lineage.** New Eng. J. Med. 316: 645-650, 1987. [PubMed: 3821794] [Full Text: https://doi.org/10.1056/NEJM198703123161101]

55. Topi, G., D'Alessandro Gandolfo, L. **Inheritance of porphyria cutanea tarda: analysis of 14 cases in 5 families.** Brit. J. Derm. 97: 617-627, 1977. [PubMed: 603745] [Full Text: https://doi.org/10.1111/j.1365-2133.1977.tb14268.x]

56. Vogler, W. R., Galambos, J. T., Olansky, S. **Biochemical effects of chloroquine therapy in porphyria cutanea tarda.** Am. J. Med. 49: 316-321, 1970. [PubMed: 5455563] [Full Text: https://doi.org/10.1016/s0002-9343(70)80022-5]

57. Waldenstrom, J. **Studien ueber Porphyrie.** Acta Med. Scand. 82 (suppl.): 1-254, 1937.

58. Wang, H., Long, Q., Marty, S. D., Sassa, S., Lin, S. **A zebrafish model for hepatoerythropoietic porphyria.** Nature Genet. 20: 239-243, 1998. [PubMed: 9806541] [Full Text: https://doi.org/10.1038/3041]

59. Ziprkowski, L., Krakowski, A., Crispin, M., Szeinberg, A. **Porphyria cutanea tarda hereditaria.** Isr. J. Med. Sci. 2: 338-343, 1966. [PubMed: 5956228]

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